

ClimbAR: Collaborative Augmented Reality for Climbing Applications

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ABSTRACT

In this work, we present ClimbAR, an open-source, collaborative, real-time, augmented reality application for the Hololens 2 that allows boulderers to set climbing holds virtually to better understand and plan routes. We performed two preliminary studies: a quantitative study that assessed the system's rotational and translational alignment accuracy, and a qualitative study that used surveys and voluntary participation to evaluate the usefulness and enjoyment of the system.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed/augmented reality.

1 INTRODUCTION

With indoor climbing's growing popularity, it is fitting that new research will apply artificial intelligence (AI) and augmented reality (AR) to this industry. Multiple works have applied mobile [1] or projector-based AR [2] or tools such as computer vision [4] and machine learning [3] to climbing. To our knowledge, existing works have not examined collaborative problems in climbing with real-time, AR systems using modern head-mounted displays (HMDs), such as that of climbing route setting explored in this work.

In this work, we present: 1) SynchronizAR, a real-time, marker-less, synchronization framework for shared virtual content across multiple Hololens 2 and 2) ClimbAR, a method of collaborative interaction for route setting using AR. SynchronizAR is the core collaborative synchronization framework for ClimbAR and is a proxy for it in our analysis.

In two informal studies, we perform a quantitative analysis of SynchronizAR and qualitative analysis of ClimbAR. In the quantitative study, we measure the translational and rotational alignment error achievable by ClimbAR via SynchronizAR. We explore whether these errors tend to increase away from the center line of the users' optical view of the wall. In the qualitative study, we gather subjective feedback from participants when using ClimbAR.

2 METHODS

Collaborative AR environments require real-time action synchronization, split into two parts: spatial alignment and movement synchronization. Also, a collaborative AR application must afford interaction cues corresponding to users' actions. In this work, we achieve spatial alignment by combining the World Locking Tools (WLT) by Microsoft and the Hololens 2's Spatial Mapping to blend the users' physical and virtual environments. To perform movement synchronization, relevant data about each virtual object, such as its position and rotation, are sent between users' HMDs via a wireless network leveraging Photon Unity Networking 2 (PUN2).

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SynchronizAR provides the core synchronization and cues common to both it and ClimbAR; synchronizing arbitrary objects across two or more HMDs. ClimbAR extends SynchronizAR to allow users to construct, save and load virtual routes. A Raspberry Pi Model 4B hosted a wireless server managing user-created routes over a local network created by a laptop and 5G cellular hotspot.

As shown in Fig. 1, both SynchronizAR and ClimbAR provide the user: 1) ray pointers from each hand, 2) virtual content color change upon user ray pointer hover, and 3) interactive sounds to give contextual cues. For multiple users all will see the effect. Application state is also conveyed to each user e.g. the number of users and connection details.

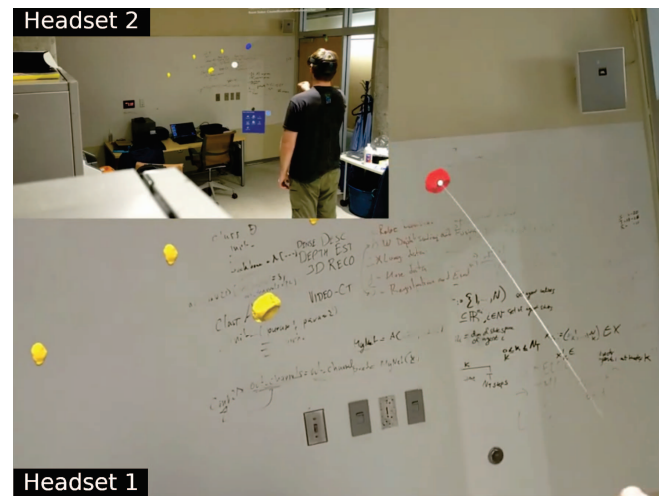


Figure 1: Synchronized climbing holds across two HMDs showing color cues on hover: HMD 1 Lower Right, HMD 2 Upper Left

Quantitative Evaluation: We evaluate the alignment accuracy of SynchronizAR (and by proxy, ClimbAR) using the Static Physical Content Method (SPCM). SPCM does not require an external camera to track the markers because the tracked markers are virtual and the application can record the marker positions directly. A total of 11 teams (each with two participants) were asked to align virtual markers (synchronized through the application and visible to both participants) with similarly sized physical ArUco markers (Fig. 2b). The physical content was positioned randomly and approximately uniformly in advance of all user study pairs and were left in place throughout the study. Team members took turns aligning the virtual markers to the physical markers (each member placing all virtual markers before the other began). The participants stood ≥ 2.5 meters from the wall and the section of the wall allowed for marker placement was 2.7 meters in width by 2.0 meters in height. The wall area qualitatively approximated a typical route-setting area for a bouldering problem at a climbing gym. This was desirable since this evaluation of SynchronizAR served as a proxy evaluation for ClimbAR.

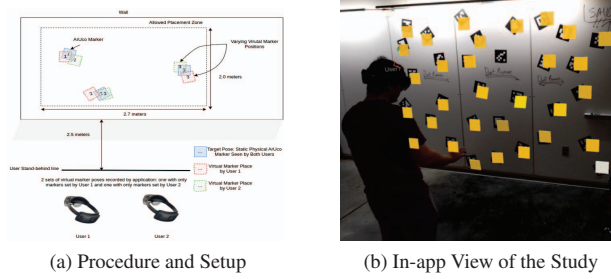


Figure 2: SPCM

Qualitative Evaluation: We performed an informal, qualitative user study of ClimbAR at two climbing gyms to evaluate its effectiveness for collaborative route setting. 14 people participated. They were asked to virtually construct routes using ClimbAR from common climbing maneuvers and were given an optional questionnaire to evaluate their experience.

3 RESULTS

Quantitative Evaluation: The 11 teams placed a total of 345 virtual markers. Each team was instructed to place 32 markers, however some teams inadvertently omitted some markers. We calculated the translational alignment error (X, Y and Z components) by averaging the euclidean distances between all marker pairs and participants. The average translational alignment error by X, Y, Z dimensions were: 2.88 cm, 0.98 cm and 1.83 cm, respectively. For any marker pair, we also calculated the rotational error between their respective orientations by: 1) converting the orientation to rotation matrices R_1 and R_2 (for User 1 and User 2 respectively), 2) computing the rotation matrix between R_1 and R_2 as $R_{12} = R_1^T R_2$, 3) computing the rotational error between R_1 and R_2 as $\theta_{error} = \arccos((\text{Trace}(R_{12}) - 1)/2)$. We then averaged $|\theta_{error}|$ across all marker pairs and participants. The average total translational alignment error was 3.85 cm and the average rotational alignment error was 12.83 degrees. Fig. 3 shows the translational and rotational alignment errors' location and magnitude in the reference frame of User 1 (among all teams) in the study.

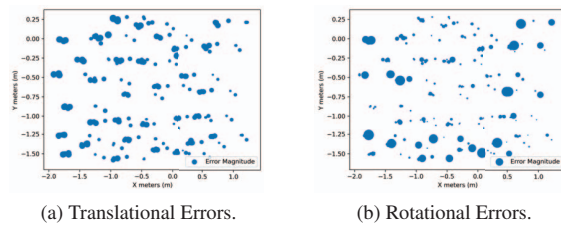


Figure 3: SPCM Alignment Errors versus ArUco Marker Location in User 1 Reference Frame. Larger dots indicate larger errors.

Qualitative Evaluation: Of the 14 questionnaire respondents, two were climbing coaches, six were regular climbers and the rest abstained from answering. When asked: What did you like? 10 responded that they enjoyed the interactive setting with AR, the ability to add holds, the user-friendly interface, intuitive, controls, multi-player functionality and futuristic feel. When asked: What did you dislike? Seven of nine respondents indicated minor issues with the pinch and place controls, either being non-intuitive or unresponsive. When asked: What other uses do you see for something like this

in climbing? eight responded, offering climb plotting and surveys, competition and training route setting, virtual gym construction, building full scale walls, setting up workouts and sharing via social media.

4 DISCUSSION AND FUTURE WORK

Quantitative Evaluation: Our results do not statistically support the idea that translational and rotational alignment error tend to increase with increasing marker-centroid distance. Using Spearman's correlation coefficient test and probability plots, at best, there is a negligible to weak, positive correlation between marker alignment errors and marker-centroid distance.

Because rotation of virtual content was not maintained during translation, users were often forced to repeat rotations once they had translated the virtual marker to the location of the physical marker. Anecdotally, some users reported fatigue due to this issue. It would be interesting to determine how much of any existing correlation between alignment error and marker-centroid distance is dependent upon the application's lack of rotation preservation versus general degradation of user perception for far-away, peripheral objects.

Moving forward, future works could compare teams to individual users for the alignment task and the maximum user limit of SynchronizAR. Future analyses could also extend to other wall geometries to determine if SPCM translation and rotational alignment errors represent a reasonable estimate of the errors expected using the application on similar collaborative tasks.

Qualitative Evaluation: ClimbAR participant feedback was positive with only minor user interface inconveniences, suggesting collaborative AR enriches the bouldering experience. Additional studies are needed to formalize and expand these preliminary results.

Future potential uses for AR in the bouldering setting include: 1) reconstruction of real-world routes within ClimbAR, 2) route problem solving, construction and quick route assessment in competitive climbing, 3) extension of ClimbAR to support beta maps (specialized notation for solving climbing problems), 4) virtual coaching 5) social media and gaming features.

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